

Maths : Combinatorics Basics & Prime Numbers

- DSA: Maths: Combinatorics Basics & Prime Numbers
- DSA: Lab Session on Prime Numbers & 2 Pointers
- DSA: Lab Session on Maths & 2 Pointers
- DSA: Backtracking
- DSA: Lab Session on Backtracking
- DSA: Linked List: Sorting and Problems
- DSA: Linked List: Doubly Linked List & Detecting Loop
- DSA: Trees 4: Morris Inorder Traversal + LCA
- DSA: Lab Session on Binary Trees 2
- DSA: Hashing 3: Internal Implementation & Problems [TreeMap & TreeSet]
- Contest 3: Math, Two Pointers, Backtracking, Linked List & Trees
- DSA: Heaps: Introduction
- DSA: Heap Sort & Greedy
- DSA: Lab Session on Heaps & Greedy
- DSA: Lab Session on Interview Problems 1
- DSA: DP 1: One Dimensional
- DSA: DP 2: Two Dimensional
- DSA: DP 3: Knapsack
- DSA: Lab Session on Applications of Knapsack
- DSA: Graphs 1: Introduction, DFS & Cycle Detection
- DSA: Graphs 2: BFS & MST
- DSA: Graphs 3: Dijkstra Algo & Topological Sort
- DSA: Lab Session on Interview Problems 2
- DSA: Revision of DSA 3 & 4
- DSA: Contest 4: Heaps, Greedy, DP & Graphs
- Contest 5: Full Syllabus

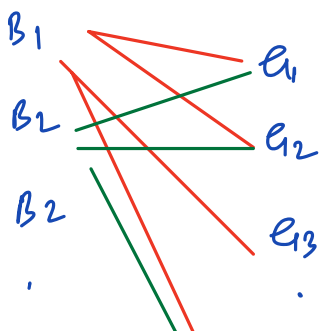
Agenda

- Intro : Addition & Multiplication Rule
- Permutations & Combinations Basics - ${}^n P_r$ or ${}^n C_r$
- Properties of ${}^n C_r$
- Prime no.
- All prime no. from 1 to N (seine)

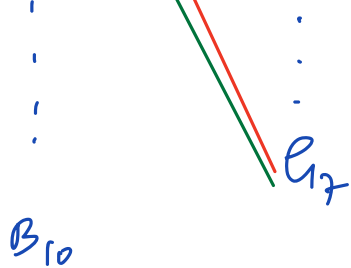
Question

Given 10 boys & 7 girls. How many pairs can be formed?

1 pair : 1 boy + 1 girl

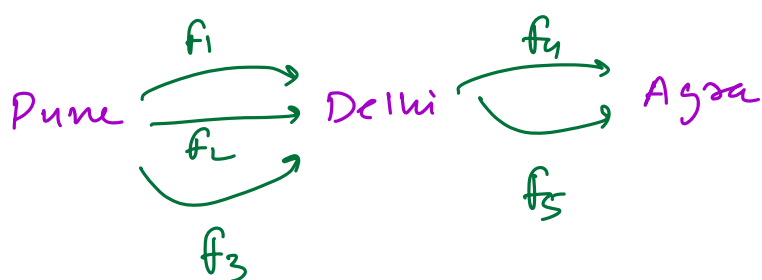


$B_1 \rightarrow 7 \text{ pairs } (G_1 \text{ to } G_7)$
 $B_2 \rightarrow 7 \text{ pairs}$
.
.
.



total pairs = $10 \times 7 = 70$ ways (pairs)

Question 2



no. of ways to travel from Pune to Agra via Delhi?

Pune $\rightarrow$ Delhi AND Delhi $\rightarrow$ Agra

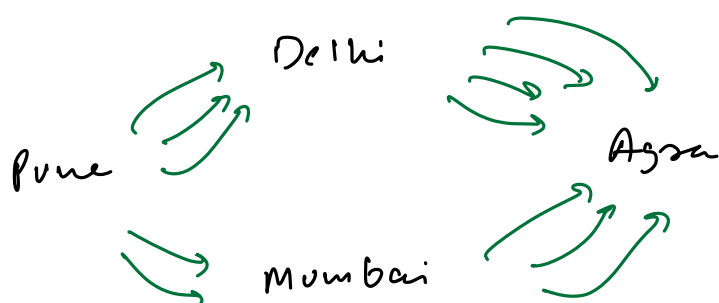
ways (Pune $\rightarrow$ Delhi) = 3

ways (Delhi $\rightarrow$ Agra) = 2

total ways : ways (Pune $\rightarrow$ Delhi) $\times$ ways (Delhi $\rightarrow$ Agra)
 $= 3 \times 2 = 6$

ways = $f_1 f_4$ $f_2 f_4$ $f_3 f_4$
 $f_1 f_5$ $f_2 f_5$ $f_3 f_5$

Quiz



Pune $\rightarrow$ Delhi AND Delhi $\rightarrow$ Agra

Pune $\rightarrow$ Agra $\Rightarrow$

OR

Pune $\rightarrow$ Mumbai AND Mumbai $\rightarrow$ Agra

ways (Pune $\rightarrow$ Delhi) = 3

ways (Delhi $\rightarrow$ Agra) = 4

ways (Pune $\rightarrow$ Delhi AND Delhi $\rightarrow$ Agra) = $3 \times 4 = 12$

$$\text{ways (Pune} \rightarrow \text{Mumbai)} = 2$$

$$\text{ways (Mumbai} \rightarrow \text{Agra)} = 3$$

$$\text{ways (Pune} \rightarrow \text{Mumbai AND Mumbai} \rightarrow \text{Agra)} = 2 \times 3 = 6$$

$$\text{total ways from Pune to Agra} = 12 + 6 = 18$$

- Multiplication $\Rightarrow$ AND : used when counting possibilities that occur together in sequence.

- Addition $\Rightarrow$ OR : used when counting possibilities that occur in separate ways

Question

find most variety of meal combos.

Given $A(n)[3]$

$A[i][0] \rightarrow$ no. of main courses

$A[i][1] \rightarrow$ no. of desserts

$A[i][2] \rightarrow$ no. of beverages

$$A = \left[\begin{array}{l} [3, 2, 2], \\ [4, 3, 3], \\ [1, 1, 1] \end{array} \right] \quad \begin{array}{l} \Rightarrow 3 \times 2 \times 2 = 12 \\ \Rightarrow 4 \times 3 \times 3 = 36 \\ \Rightarrow 1 \times 1 \times 1 = 1 \end{array}$$

Solution: Iterate the array & find the max combo restaurant.

Permutations

Arrangement of objects

order matters $\Rightarrow (i, j) \neq (j, i)$

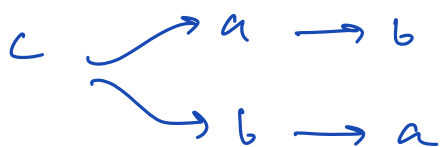
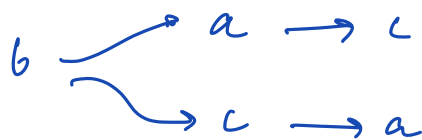
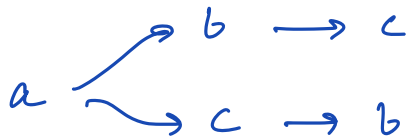
Ques: # ways to arrange 3 distinct characters

$a, b, c \Rightarrow$

abc	bac	cab
acb	bca	cba

ans = 6

$$\underline{3} \times \underline{2} \times \underline{1} = 6 \quad (3!)$$



for 4 distinct characters $\rightarrow 4! \quad (4 \times 3 \times 2 \times 1 = 24)$

ways to arrange N distinct characters?

$$N \times N-1 \times N-2 \times \dots \times 1 = N!$$

Question

find # ways to arrange R out of N distinct chars.

d, a, t, e

$$\underline{4} \times \underline{3} = 12$$

$$N=4, R=2$$

da	ad	td	ed
dt	at	ta	ea
de	ae	te	et

Select R out of N

1 2 3 . - , , , R

$$\# \text{ ways} \rightarrow N \quad N-1 \quad N-2 \quad \dots \quad N-(R-1) \\ = N-R+1$$

$$N \times (N-1) \times (N-2) \times \dots \times (N-R+1) \times (N-R) \times (N-R-1) \times \dots \times 1$$

$$(N-R) \times (N-R-1) \times \dots \times 1$$

$$\Rightarrow \frac{N!}{(N-R)!}$$

$${}_N P_R = \frac{N!}{(N-R)!}$$

$$R \leq N$$


 # ways to arrange R elements
 out of total N elements

Combinations

selection of objects

order doesn't matter $\Rightarrow (i,j) = (j,i)$

Ques : Given 4 players , count ways to select 3 players.

$\{ P_1, P_2, P_3, P_4 \}$

P_1, P_2, P_3

P_1, P_2, P_4

P_1, P_3, P_4

P_2, P_3, P_4

} 4 ways

Q₄₄: Given 4 players, write ways to arrange players in 3 slots.

$\{P_1, P_2, P_3, P_4\}$

P_1, P_2, P_3
 P_1, P_3, P_2
 P_2, P_1, P_3
 P_2, P_3, P_1
 P_3, P_1, P_2
 P_3, P_2, P_1

P_1, P_2, P_4
 P_1, P_4, P_2
 P_2, P_1, P_4
 P_2, P_4, P_1
 P_4, P_1, P_2
 P_4, P_2, P_1

P_1, P_3, P_4
 P_1, P_4, P_3
 P_3, P_1, P_4
 P_3, P_4, P_1
 P_4, P_1, P_3
 P_4, P_3, P_1

P_2, P_3, P_4
 P_2, P_4, P_3
 P_3, P_2, P_4
 P_3, P_4, P_2
 P_4, P_2, P_3
 P_4, P_3, P_2

$\{P_1, P_2, P_3\}$

$\{P_1, P_2, P_4\}$

$\{P_1, P_3, P_4\}$

$\{P_2, P_3, P_4\}$

4 ways to select 3 players

$${}^4P_3 = \frac{4!}{(4-3)!} = \frac{4!}{1!} = 24 \text{ ways to arrange 3 players}$$

selection AND arrangement = permutation

$$4 \times 3! = 4 \times 6 = 24$$

select R elements out of N elements

$${}^N C_R \times R! = {}^N P_R$$

$${}^N C_R = \frac{{}^N P_R}{R!} = \frac{N!}{R! \times (N-R)!}$$

Properties of ${}^n C_r$

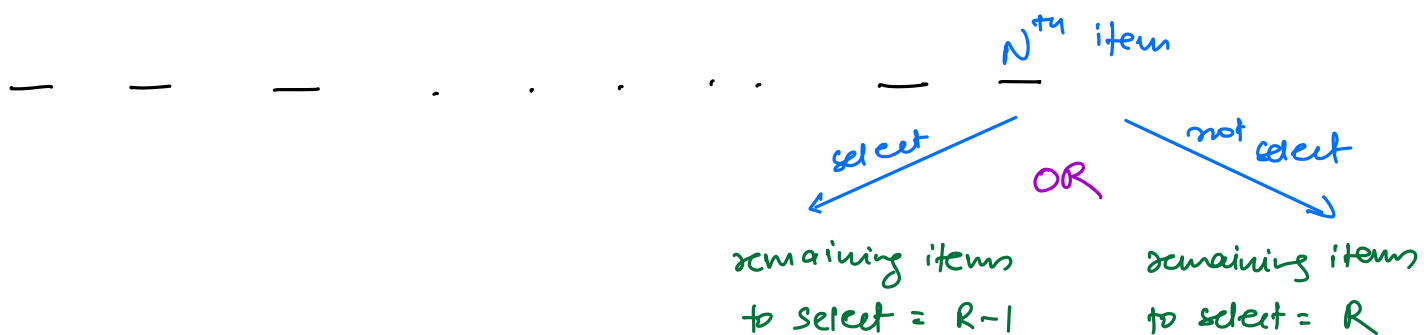
$$\begin{aligned} 1. \text{ \# ways to not select anything} &= {}^N C_0 = \frac{N!}{0! (N-0)!} \\ &= \frac{N!}{N!} = 1 \end{aligned}$$

$$\begin{aligned}
 2. \text{ \# ways to select everything} &= {}^N C_N = \frac{N!}{N! (N-N)!} \\
 &= \frac{N!}{N!} = 1
 \end{aligned}$$

$$\begin{aligned}
 3. \text{ \# ways to select } (N-R) \text{ items} &= {}^N C_{N-R} \\
 &= \frac{N!}{(N-R)! \times (N-(N-R))!} \\
 &= \frac{N!}{(N-R)! \times R!} = {}^N C_R
 \end{aligned}$$

$${}^N C_R = {}^N C_{N-R}$$

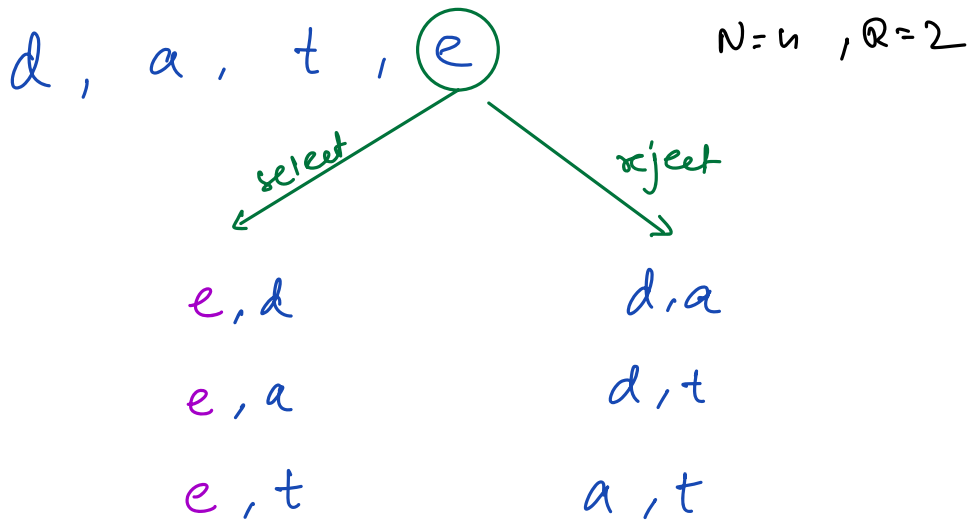
4. \# ways to select R items from N items



total remaining
item = N-1

total remaining
item = N-1

$${}^N C_R = {}^{N-1} C_{R-1} + {}^{N-1} C_R$$



$${}^4 C_2 = \frac{4!}{2! (4-2)!}$$

$$= \frac{4 \times 3 \times 2}{2 \times 2} = 6$$

$${}^{N-1} C_{R-1} = {}^3 C_1 = 3$$

$${}^{N-1} C_R = {}^3 C_2 = 3$$

$$3+3=6$$

Prime Numbers : Numbers that have exactly 2 factors

ex: 2, 3, 5, 7, ...

Given a number, check if it is prime?

Solution : count # factors

Code

```
bool checkPrime(N) {  
    count = 0  
    for (i = 1 to  $\sqrt{N}$ ) {  
        if (N % i == 0) {  
            if (i == N/i) count++;  
            else count += 2;  
        }  
    }  
    return (count == 2);  
}
```

TC = $O(\sqrt{N})$

SC = $O(1)$

Question: Generate prime no. from 1 to N.

N=10 $\Rightarrow$ 2, 3, 5, 7

N=20 $\Rightarrow$ 2, 3, 5, 7, 11, 13, 17, 19

Brute force

```
for (i=2 to N) {
```

```
    if (checkPrime(i)) {
```

```
        print(i)
```

```
    }  
}
```

TC = $O(N\sqrt{N})$

SC = $O(1)$

Sieve of Eratosthenes

- Initially, every number is prime
- for every prime no, make its multiples non-prime.

✓ 2	✓ 3	4	✓ 5	6	✓ 7	8	9	10	✓ 11	12	✓ 13	14...
T	T	T	T	T	T	T	T	T	T	T	T	T ...

$5 \times 2 = 10$
 $5 \times 3 = 15$
 $5 \times 4 = 20$
 $5 \times 5 = 25$
 $\vdots$
 $\vdots$

code

```

bool prime[n+1]
if (prime[i]) = true
prime[1] = false

```

```

for (i = 2 to  $\sqrt{N}$ ) {
    if (prime[i]) {

```

```

        for (j = i*i; j <= n; j += i) {
            prime[j] = false

```

```

        }
    }

```

```

}

```

multiples of i

$i, 2i, 3i, 4i, 5i, \dots$


$\rightarrow$ loop will not run for $i > \sqrt{N}$

```
for (i = 2 to N) {
```

```
    if (prime[i])
```

```
        print(i)
```

```
}
```

Time Complexity

i	j	iterations
2	4, 6, 8, 10,	$\sim N/2$
3	9, 12, 15,	$\sim N/3$
4	—	0
5	25, 30, 35,	$\sim N/5$
.		+
.		.
.		.
.		.
.		+
$\sqrt{N}$		$\sim N/\sqrt{N}$

$$\text{total iterations} = \frac{N}{2} + \frac{N}{3} + \frac{N}{5} + \frac{N}{7} + \dots$$

$$= N \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \dots \right)$$

$\downarrow$
 $\log(\log N)$

$$TC = O(N \log(\log N)) \\ \approx O(N)$$

$$SC = O(N)$$

$$N = 10^{12}$$

$$\log N = 40$$

↓

$$2^{10} = 10^3$$

$$2^{40} = 10^{12}$$

$$\log(\log N) = \log(40) : \sim 6$$

$$\sqrt{N} = 10^6$$